

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices,
transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 40%

QUESTIONS: 10

Total Marks:50

Annexure A

Coding challenge

Code of Conduct:

*I C KARTHIK REDDY(192425157)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:*C KARTHIK REDDY**

Annexure A

Coding challenge

1. Write a Python script that generates a star topology diagram (ASCII or graphical) given n hosts and a central switch. The script must print each host label, the switch name, and the connecting Cat6 UTP cable length.

```
EXP1.py - C:/Users/ckart/Downloads/EXP1.py (3.11.9)
File Edit Format Run Options Window Help
n = int(input("Enter number of hosts: "))
switch = "SW1"

print("\nStar Topology")
print("      " + switch)

for i in range(1, n + 1):
    length = int(input(f"Enter cable length for H{i} (m): "))
    print(f"      |--- Cat6 UTP {length}m ---> H{i}")

|

IDLE Shell 3.11.9
File Edit Shell Debug Options Window Help
Python 3.11.9 (tags/v3.11.9:de54cf5, Apr  2 2024, 10:12:12) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/ckart/Downloads/EXP1.py =====
Enter number of hosts: 3

Star Topology
SW1
Enter cable length for H1 (m): 20
|--- Cat6 UTP 20m ---> H1
Enter cable length for H2 (m): 30
|--- Cat6 UTP 30m ---> H2
Enter cable length for H3 (m): 25
|--- Cat6 UTP 25m ---> H3
>>>
```

2. Code a function `validate\_segment(length\_m)` that returns True if the UTP segment is within IEEE 802.3 limits (≤ 100 m) and False with an error message otherwise. Include at least 3 test cases.

```
EXP2.py - C:/Users/ckart/Downloads/EXP2.py (3.11.9)
File Edit Format Run Options Window Help
def validate_segment(length_m):
    if length_m <= 100:
        return True
    print("Error: UTP segment exceeds 100 m")
    return False

print(validate_segment(50))
print(validate_segment(100))
print(validate_segment(120))
|
```

```
==== RESTART: C:/Users/ckart/Downloads/EXP2.py =====
True
True
Error: UTP segment exceeds 100 m
False
>>> |
```

3. Write a script that simulates CSMA/CD on a star topology with 4 hosts. Log every collision detection event and the back-off interval to a file named `csma\_log.txt`.

```
EXP3.1.py - C:/Users/ckart/AppData/Local/Programs/Python/Python311/EXP3.1.py (3.11.9)
File Edit Format Run Options Window Help
import random

hosts = ["H1", "H2", "H3", "H4"]

with open("csma_log.txt", "w") as f:
    for i in range(5):
        h1, h2 = random.sample(hosts, 2)
        f.write(f"Collision detected: {h1}, {h2}\n")

        backoff = random.randint(1, 10)
        f.write(f"Back-off interval: {backoff}\n")

print("CSMA/CD simulation completed.")
print("Log saved in csma_log.txt")
|

>>> ===== RESTART: C:/Users/ckart/AppData/Local/Programs/Python/Python311/EXP3.1.py =====
CSMA/CD simulation completed.
Log saved in csma_log.txt
>>> |
```

4. Using Python's 'socket' library, write a client-server program that models two hosts communicating through a star switch. Demonstrate message forwarding by printing the switch's MAC address table after each frame transmission.

```
EXP4.1.py - C:/Users/ckart/AppData/Local/Programs/Python/Python311/EXP4.1.py (3.11.9)
File Edit Format Run Options Window Help

import socket

s = socket.socket()
s.bind(("localhost", 5000))
s.listen(1)

print("Switch MAC: AA:BB:CC:DD:EE:FF")
print("MAC Table: H1 -> 11:22:33:44:55:66")

c, addr = s.accept()
msg = c.recv(1024).decode()

print("Frame received:", msg)
print("Forwarding frame to H2")
print("MAC Table: H1 -> 11:22:33:44:55:66, H2 -> 66:55:44:33:22:11")

c.close()
s.close()
|
```

```
client.py - C:/Users/ckart/AppData/Local/Programs/Python/Python311/client.py (3.11.9)
File Edit Format Run Options Window Help

import socket

s = socket.socket()
s.connect(("localhost", 5000))
s.send(b"Hello from H1 to H2")
s.close()
```

```
===== RESTART: C:/Users/ckart/AppData/Local/Programs/Python/Python311/EXP4.1.py =====
Switch MAC: AA:BB:CC:DD:EE:FF
Waiting for H1...

Frame received from H1: Hello from H1 to H2
Forwarding frame to H2
MAC Table:
H1 -> 11:22:33:44:55:66
H2 -> 66:55:44:33:22:11
>>>|
```

5. Create a class `StarSwitch` with methods `add\_port(host\_id)`, `remove\_port(host\_id)`, and `broadcast(frame)`. Broadcast must send a frame to all ports except the source and log each delivery.

```
EXP5.py - C:/Users/ckart/AppData/Local/Programs/Python/Python311/EXP5.py (3.11.9)
File Edit Format Run Options Window Help

class StarSwitch:
    def __init__(self):
        self.ports = []

    def add_port(self, host_id):
        self.ports.append(host_id)

    def remove_port(self, host_id):
        self.ports.remove(host_id)

    def broadcast(self, frame):
        source = frame["source"]

        for host in self.ports:
            if host != source:
                print(f"Frame delivered to {host}")

sw = StarSwitch()

sw.add_port("H1")
sw.add_port("H2")
sw.add_port("H3")

sw.broadcast({"source": "H1", "data": "Hello"})

>>>
===== RESTART: C:/Users/ckart/AppData/Local/Programs/Python/Python311/EXP5.py =====
Frame delivered to H2
Frame delivered to H3
>>>
```